Analysis and Design of Passive Components for Interleaved Flyback Converter with Integrated Transformer

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***Abstract*— Flyback switch mode power supplies have been widely used in low-power applications, such as DC/DC converters, solar micro-inverters and LED drivers. However, flyback converters have several problems related to the volume of the output capacitor as well as high output voltage noise; this is produced by the discontinuous output current. Consequently, interleaved operation with parallel connection on the secondary side can reduce the output current ripple compared with the single-phase flyback converter. Nevertheless, besides the output capacitor, the interleaved operation is unsuitable for transformers if it is desired to obtain light weight and compact performance. To address this problem, the interleaved flyback converter with integrated transformer has been proposed for achieving miniaturization of the output capacitor and transformers. There are mainly two types of the proposed flyback converter with integrated transformer: Parallel and Series types. These two types are categorized based on the connection on the primary side. Nevertheless, quantitative comparison of the volume and power loss has not been analyzed yet. Moreover, the design method of the integrated transformer also has not been conducted with clarity. Hence, in this paper, in order to provide a guide for the design of an interleaved flyback converter for achieving high power density, we analyze the quantitative volume and power losses of the integrated transformer and the input capacitor. Finally, this paper shows some experimental results that validate the appropriateness of the design method for the integrated transformer.**

Keywords—Flyback converter; integrated transformer; input capacitor ; parallel type; series type

I. INTRODUCTION

Flyback converters are widely used in switching mode power supplies in order to provide regulated output voltages for low-power applications such as LED drivers, solar micro inverters, and DC-DC converters [1]-[5]. In these applications, high power density is required to prevent global warming and depletion of energy resources [6]-[7]. To address this requirement, interleaved operation has been introduced because of its several features: This topology offers a reduction of the filter size because the interleaved operation of the two cells reduces the current ripple in the output capacitor: this method can improve the rated power under the Discontinuous Conduction Mode (DCM) because the peak and RMS currents can be reduced [1]-[2]. However, this topology presents the drawback of increasing volume and weight in the magnetic components in comparison with the single phase flyback converter. The volume and weight of the magnetic components significant influence on the cost of the flyback switch mode power supplies because magnetic components are made of expensive metals (Fe, Al, Cu, etc.) and rare materials (Ni, Mn, Co etc.) [6]. As a consequence, interleaved converters with integrated transformers have been proposed in [5]-[6]. This technique offers the following features [6]-[8]:

- Higher frequency operation in the primary currents and

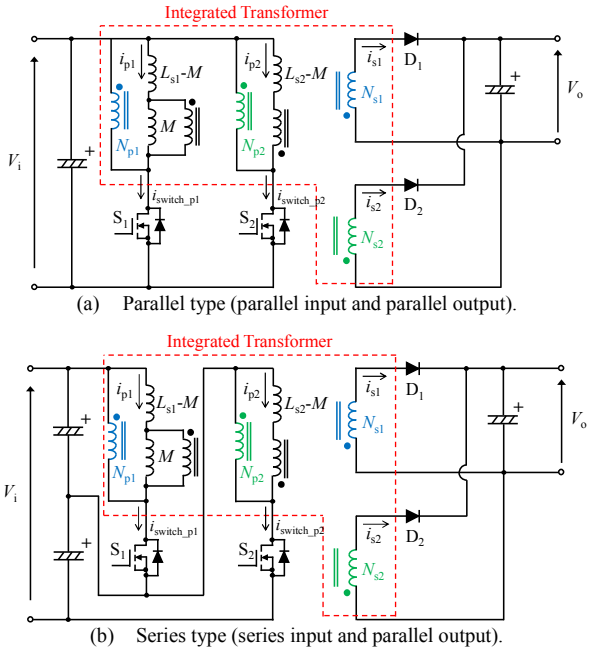


Fig. 1. Interleaved Flyback Converter with Integrated Transformer.

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secondary currents can be achieved by the mutual induction effect. Therefore, the inductance required to produce a determined current ripple will be reduced.

- Lower iron and copper losses than converters without integrated transformers because each inductor or transformer shares its magnetic path and windings. Consequently, the size of the core and windings is reduced.

Furthermore, there are mainly two types of the proposed flyback converter with integrated transformer. The first one is the parallel input parallel output interleaved flyback converter (parallel type) shown in Fig. 1 (a); and the second one is the series input parallel output interleaved flyback converter (series type) shown in Fig. 1 (b) [6]. The series type is able to reduce the voltage stresses of the primary side, and the parallel type is able to reduce the current stresses of the primary side. In general, the circuit topology had been selected by the rated parameters of the power semiconductors. However, recently, with the evolution of the power semiconductors, the selection of the circuit topology is becoming unrestrained [9]-[10]. For this reason, passive components such as integrated transformers and input capacitors must be considered as well. However, in terms of high power density performance under the same conditions, a quantitative comparison has not been investigated yet.

Hence, in this paper, we analyze the electromagnetic characteristics of the integrated transformer and the input capacitor. Moreover, this paper provides the design guidelines of these two topologies as well as the discussion of volume and power losses. Finally, this paper shows the experimental validation, where the design method for integrated transformers is validated.

II. Analysis and Design of The Integrated Transformer

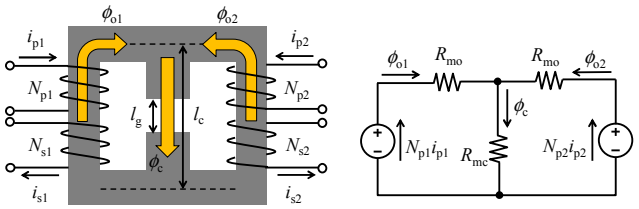
A. Design Method

In this section, we discuss the design method for the integrated transformer, which uses a generic core of a known shape. In this and the following sections, the converter operation is discussed under the Continuous Conduction Mode (CCM). In addition, in order to have an analysis as simple as possible, it is assumed that the leakage inductances are much smaller than the self-inductance, and all the switches and diodes are ideal. Finally, assuming that the efficiency of the power converter is 100%, the output power is expressed as follows:

$$P_o = I_o \cdot V_o = I_i \cdot V_i \tag{1}$$

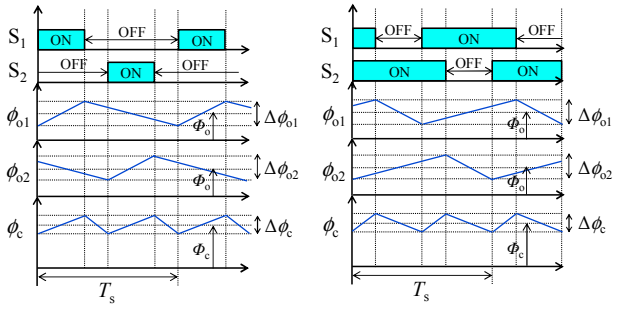
where P_o is the output power, I_o is the output current, and I_i is the input current. Moreover, for the following discussions, the parallel type topology is designed and analyzed in terms of the variables shown in the equation (1). On the other hands, in case of the series type, the input voltage V_i is replaced with the half of the value of the parallel type and, the input current I_o is replaced with twice the value of the parallel type.

The magnetic core structure of an interleaved flyback converter, when the transformer is made with an EE or EI core, is shown in Fig. 2-(a). And, Fig. 2-(b) shows the magnetic circuit model of Fig. 2-(a). In order to design the integrated transformer, it is necessary to determine the number of winding turns and the gap length. First, we calculate the number of winding turns. Therefore, the magnetic components required to keep the transformer without magnetic saturation are taken into account. In the integrated transformer, the characteristics of the peak fluxes in the outer legs and in the central leg are different, see Fig. 3. In the case of a duty ratio d<0.5 and d>0.5, the peak fluxes ϕ_{op} in the outer legs and the peak fluxes ϕ_{cp} in the central leg are determined by the equations (2)-(4) of Table I. Where T_s is the switching period, N_p is the number of winding turns of the primary side, R_{mc} is the magnetic reluctance of the central leg, R_{mo} is the magnetic reluctance of outer leg, and β is the ratio of number of turns ($\beta=N_s/N_p$). The first term on the right of the equations (2)-(5) indicates the DC fluxes Φ_o, Φ_c caused by the DC bias current. The second term on the right of the equations (2)-(4) indicates the AC fluxes $\Delta\phi_{o1}$, $\Delta\phi_{o2}$, and $\Delta\phi_c$ calculated with the Faraday's law. In addition, for the case of d>0.5 and d<0.5, the peak-to-peak ripple current Δi_p, necessary for the design of the integrated transformer, is determined by the equations (5)-(6) of Table I.



(a) Magnetic structure (b) Magnetic circuit model

Fig. 2. Integrated Transformer.



(a) d<0.5 (b) d>0.5

Fig. 3. Magnetic flux waveforms of each phase.

According to the study reported in [11], the maximum flux of the central leg does not become twice or more the maximum flux of the outer leg in all the values of duty ratio d. Therefore, in case of the design of the integrated transformer, if we assume that the sectional area in the central leg is more than twice as large as the sectional area in the outer leg, we only should consider the flux density of the outer legs. The following conditional equation is obtained from the condition that peak fluxes ϕ_{op} in the outer legs do not exceed the maximum flux ϕ_{omax}:

$$\phi_{op} < \phi_{o\max} \tag{7}$$

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TABLE I. RESULTS OF THE ELECTROMAGNETIC ANALYSIS

Dury range	Peak magnetic fluxes ϕ_{op} in outer leg	Peak magnetic fluxes ϕ_{cp} in outer leg
d<0.5	$\frac{N_p}{R_{mo}+2R_{mc}}\cdot\frac{I_i+\beta\cdot I_o}{2}+\frac{1}{2}\cdot\frac{V_i}{N_p}\cdot d\cdot T$ (2)	$\frac{N_p\cdot(I_i+\beta\cdot I_o)}{R_{mo}+2R_{mc}}+\frac{V_i}{N_p}\cdot\frac{d}{1-d}\cdot\left(\frac{1}{2}-d\right)\cdot T_s$ (3)
d>0.5		$\frac{N_p\cdot(I_i+\beta\cdot I_o)}{R_{mo}+2R_{mc}}+\frac{V_i}{N_p}\cdot\left(d-\frac{1}{2}\right)\cdot T_s$ (4)
Duty range	Peak-to-peak ripple currents Δi_p of each phase	
d<0.5	$\frac{V_i}{N_p^2}\cdot\left(R_{mo}+\frac{1-2d}{1-d}\cdot R_{mc}\right)\cdot d\cdot T_s$ (5)	
d>0.5	$\frac{V_i}{N_p^2}\cdot\left(R_{mo}-\frac{1-2d}{d}\cdot R_{mc}\right)\cdot d\cdot T_s$ (6)	

In the case of d<0.5, the number of turns of the primary side N_p can be calculated by substituting (2) into (7) as follows:

$$2\frac{1-d}{1-2d}\cdot\frac{\Delta i_p\cdot\phi_{o\max}}{V_i\cdot d\cdot T_s}\cdot N_p^3-\left(\frac{I_i+\beta\cdot I_o}{2}+\frac{1-d}{1-2d}\cdot\Delta i_p\right)\cdot N_p^2+\left(1-2\frac{1-d}{1-2d}\right)\cdot R_{mo}\cdot\phi_{o\max}\cdot N_p+\left(\frac{1-d}{1-2d}-\frac{1}{2}\right)\cdot R_{mo}\cdot V_i\cdot d\cdot T_s>0 \quad (8)$$

The magnetic reluctance of the central leg R_{mc} is given by the substituting of N_p into (5) as follows:

$$R_{mc}=\frac{1-d}{1-2d}\cdot\left(\frac{\Delta i_p\cdot N_p^2}{V_i\cdot d\cdot T_s}-R_{mo}\right) \quad (9)$$

In the case of d>0.5, similarly, the number of turns of the primary side N_p can be calculated by substituting (2) into (7) as the following equation:

$$\frac{2d}{2d-1}\cdot\frac{\Delta i_p\cdot\phi_{o\max}}{V_i\cdot d\cdot T_s}\cdot N_p^3-\left(\frac{I_i+\beta\cdot I_o}{2}+\frac{d}{2d-1}\cdot\Delta i_p\right)\cdot N_p^2+\left(1-\frac{2d}{2d-1}\right)\cdot R_{mo}\cdot\phi_{o\max}\cdot N_p+\left(\frac{d}{2d-1}-\frac{1}{2}\right)\cdot R_{mo}\cdot V_i\cdot d\cdot T_s>0 \quad (10)$$

Also, R_{mc} is given by substituting N_p into (6) as:

$$R_{mc}=\frac{d}{1-2d}\cdot\left(R_{mo}-\frac{\Delta i_p\cdot N_p^2}{V_i\cdot d\cdot T_s}\right) \quad (11)$$

Moreover, the number of turns of the secondary side N_s is voluntarily determined by the basic input-output voltage relational expression.

Then, we can calculate the gap length l_g. The magnetic reluctance of the central leg R_{mc} is obtained on the base of Fig. 2, as follows:

$$R_{mc}=\frac{l_c-l_g}{\mu_o\cdot\mu_r\cdot A_c}-\frac{l_g}{\mu_o\cdot A_c} \quad (12)$$

Based on the equations (9), (11), and (12), the gap length l_g is determined. Where, μ_o is the permeability of the free-space, μ_r is the relative permeability, and A_c is the sectional area of the central leg.

B. *Appropriateness verification of the design method*

In this section, we verify the accuracy of the design method of the integrated transformer from the experimental tests of the parallel and series type. The verification of the accuracy of the design method can be indirectly explained by the verification of the current ripple of the primary side. The experimental parameters are shown in Table II. In this context, the prototypes are constructed with an RCD snubber circuit. The core of the both parallel and series types is the EER42-Z (material PC47 ferrite, manufactured by TDK Co., Ltd). Consequently, the experimental waveforms of the parallel type are shown in Fig. 4, where V_{ds1} and V_{ds2} is the drain-to-source voltage. Similarly, the experimental waveforms of the series type are shown in Fig. 5. In these experiments, the output voltage presents a drop due to the diode forward voltage and the leakage inductance. However, the design of the integrated transformer is not related to this issue. In fact, it is related to the output load, therefore, we adjust the load resistance to 200W. As it can be seen from these waveforms, we can confirm that the magnetic saturations do not occur in each integrated transformer. And also, the measured values of the peak-to-peak ripple currents of primary side approximate

TABLE II. SPECIFICATION FOR THE EXPERIMENTSAL PARAMETERS

Item		Parallel type	Series type
Input voltage	V_i	24 V	
Output voltage	V_o	170V	
Output power	P_o	200 W	
Duty ratio	d	0.7	
Switching frequency	f_s	100 kHz	
Ratio of the number of winding ($\beta=N_p/N_s$)	β	3	6
Primary's number of turns	N_p	9 turns	5 turns
Initial per-meability	μ_r	2300	
maximum flux density	B_{max}	350 mT	
Ripple current of the primary side	Δi_p	1.79 A	3.57 A
Self-inductance	L_s	177.4 μH	53.4 μH
Mutual inductance	M	143.2 μH	45.5 μH
Leakage inductance	L_{lk}	1.1 μH	0.84 μH

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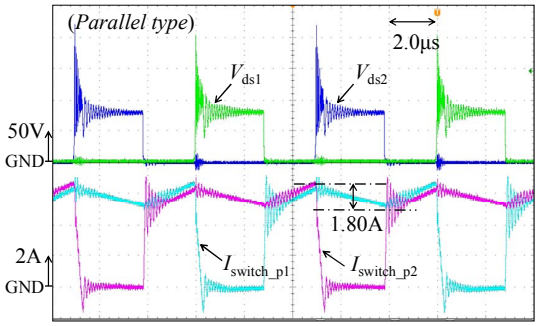


Fig. 4. Experimental waveforms of the parallel type.

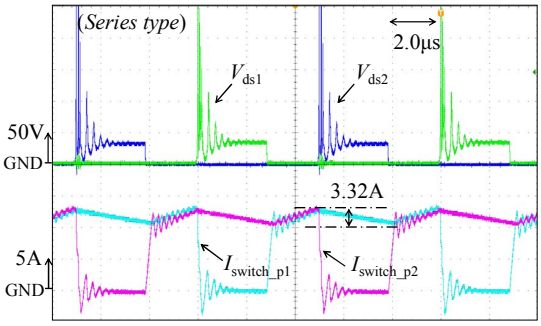


Fig. 5. Experimental waveforms of the series type.

TABLE III. RESULTS OF THE ELECTROMAGNETIC ANALYSIS OF THE HYPOTHETICAL CORE

Duty range	Peak magnetic fluxes ϕ_{op} in the outer leg	Magnetic reluctance R_{mo} in the outer leg
d<0.5	$\frac{V_i \cdot d \cdot T_s \cdot \{1-(1+k)\cdot d\}}{(1+k)\cdot(1-d)\cdot k_i \cdot N_p} + \frac{1}{2}\cdot\frac{V_i}{N_p}\cdot d \cdot T_s$ (15)	$\frac{k_i \cdot (I_i+\beta\cdot I_o)\cdot N_p{}^2 \cdot (1-d)\cdot(1-k)}{2V_i \cdot d \cdot T_s \cdot \{1-d\cdot(1+k)\}}$ (17)
d>0.5	$\frac{V_i \cdot T_s \cdot \{d\cdot(1+k)-k\}}{(1+k)\cdot k_i \cdot N_p} + \frac{1}{2}\cdot\frac{V_i}{N_p}\cdot d \cdot T_s$ (16)	$\frac{k_i \cdot (I_i+\beta\cdot I_o)\cdot N_p{}^2 \cdot (1-k)}{2V_i \cdot T_s \cdot \{d\cdot(1+k)-k\}}$ (18)
Duty range	Gap length l_g	
d<0.5	$\frac{\mu_r \cdot \mu_0 \cdot A_c}{\mu_r - 1}\cdot\left\{\frac{1-d}{2d-1}\cdot\frac{k_i\cdot(I_i+\beta\cdot I_o)\cdot N_p{}^2}{4V_i\cdot d\cdot T_s}\cdot\frac{(1-2d)\cdot k}{(1+k)\cdot d-1}\right\} - \frac{l_c}{\mu_r-1}$ (19)	
d>0.5	$\frac{\mu_r \cdot \mu_0 \cdot A_c}{\mu_r - 1}\cdot\left\{\frac{d}{1-2d}\cdot\frac{k_i\cdot(I_i+\beta\cdot I_o)\cdot N_p{}^2}{4V_i\cdot d\cdot T_s}\cdot\frac{(1-2d)\cdot k}{(1+k)\cdot d-k}\right\} - \frac{l_c}{\mu_r-1}$ (20)	

match with the designed values. Therefore, the accuracy of the design method of the magnetic components is proved from the experimental results.

C. Volume calculation model of the integrated transformer

In this section, we compare the volume of the magnetic components using the hypothetical core concept, studied in [6],[7],[12], of the integrated transformer shown in Fig.3 with the parallel type and the series type. The window shape and the cross section of the hypothetical core are defined as squares for simplicity. The window area A_w can be calculated from:

$$A_w = \frac{N_p \cdot A_{winding_p} + N_s \cdot A_{winding_s}}{k_s} \quad (13)$$

where $A_{winding_p}$ is the sectional area of the primary winding, $A_{winding_s}$ is the sectional area of the secondary winding, and k_s is the space factor. In case of the volume analysis using hypothetical cores, the magnetic reluctance R_{mc}, R_{mo} are not obvious because the magnetic reluctance is calculated from the volume and shape of the core. Therefore, the magnetic reluctances R_{mc}, R_{mo} of the above equations are replaced by the coupling coefficient k that can be freely decided by the designer using the following equation:

$$k = \frac{M}{L_s} = \frac{R_{mc}}{R_{mo} + R_{mc}} \quad (14)$$

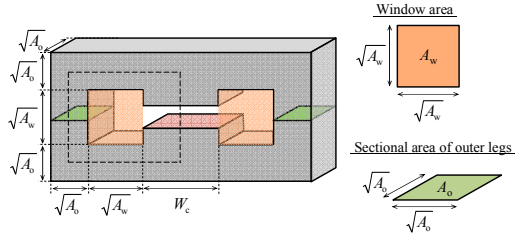


Fig. 6. Hypothetical core of the integrated transformer.

In fact, the equation (14) can be rewritten as shown in Table III. Where, k_i is the ratio of the ripple current flowing through the primary winding to the DC current as follows:

$$k_i = \frac{2\Delta i_p}{I_i + \beta \cdot I_o} \quad (21)$$

Then, the volume of the hypothetical core of the integrated core can be determined by the equations (7),(9),(11), and (15)-(20).

D. Loss calculation model of integrated transformer

In this section, the power losses of the integrated transformer are analyzed and compared between the parallel type and the series type. First, core losses are calculated using the improved Generalized Steinmetz Equation (iGSE) [13] as follows:

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$$P_v = \frac{1}{T_s} \cdot \int_0^{T_s} k_i \cdot \left|\frac{dB}{dt}\right|^{\alpha} \cdot (\Delta B)^{\beta-\alpha} \cdot dt \tag{22}$$

where

$$k_i = \frac{k}{(2\pi)^{\alpha-1} \cdot \int_0^{2\pi} |\cos\theta|^{\alpha} \cdot 2^{\beta-\alpha} \cdot d\theta} \tag{23}$$

and, the parameters k, α, and β are the Steinmetz constants, P_v is the time-average core loss per unit volume, and ΔB is the peak-to-peak flux density. The iGSE is well known as a core loss calculation method under different excitation conditions, without requiring extra characterization of the material parameters compared to the conventional Steinmetz equation [14]. This approach is widely applied in [15] and [16].

The peak-to-peak flux density ΔB and the time change rate of the flux density dB/dt are calculated by the second right-hand term of the equations (2)-(4).

Next, we calculate the copper losses of the integrated transformer. The copper losses P_{copper} can be calculated from the DC resistance R_{dc}, and AC resistance R_{ac} of the winding as shown as follows [7], [12]:

$$P_{copper} = R_{dc} \cdot {I_{Wave}}^2 + R_{ac} \cdot {I_{Wac}}^2 \tag{24}$$

where, I_{Wave} is the average current that flows into the windings, I_{Wac} is the effective value of the AC current that flows into the windings. In addition, the sectional area of the winding model is defined as square as shown in Fig 7, the length of the winding l_w, the DC resistance R_{dc}, and the AC resistance R_{ac} are calculated as follows:

$$l_W = 4\left(\sqrt{A_{winding}} + \sqrt{A_{core}}\right) \cdot N \tag{25}$$

$$R_{dc} = \rho \frac{l_W}{A_{winding}} \tag{26}$$

$$R_{ac} = \rho \frac{l_W}{4\delta \cdot \left(\sqrt{A_{winding}} - \delta\right)} \tag{27}$$

$$\delta = \sqrt{\frac{\rho}{\pi \cdot f_s \cdot \mu}} \tag{28}$$

where ρ is the electrical resistivity and δ is the skin depth of the winding.

From the analysis described above, we compare the parallel type with the series type in terms of volume and power losses. The circuit parameters are shown in Table IV. In this comparison, the volume and the losses of the magnetic components are compared under both step-up and step-down operations. The core material is the PC47 (ferrite) manufactured by TDK Co., Ltd. Fig.8 shows the relationship between the losses (sum of core and copper loss) and the volume (sum of core and winding volume) in case of changing the duty ratio d in the step-up operation. And, Fig.9 shows the results of the step-down operation. From these results, at the same value of coupling coefficient, the series type is effective for high power density of the magnetic components compared to the parallel type. In other word, in most the cases, the series type topology should be selected for integrated transformer

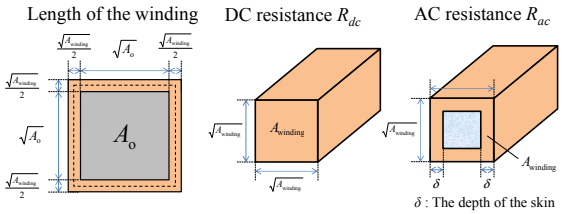


Fig. 7. Definitions of the length and the resistance of the winding.

TABLE IV. CIRCUIT PARAMETERS FOR THE COMPARISON

Parameter		Step-up	Step-down
Input voltage	V_i	24 V	400 V
Duty ratio	d	d>0.5	d<0.5
Output power	P_o	200 W	
Coupling coefficient	k	0.95, 0.90, 0.60	
Switching frequency	f_s	100 kHz	
Ratio of the number of winding of the parallel type	β_{parallel}	2.5	0.2
Ratio of the number of winding of the series type	β_{series}	5.0	0.4
Primary's number of turns	N_p	10 turns	25 turns
Space factor	k_s	0.7	
Ratio of the ripple current	k_i	0.3	
Winding current density	j	5 A/mm^2	
Electrical resistivity	ρ	1.68×10^{-8} Ωm	
Steinmetz constants		α=1.54 , β=2.39, k=0.378, k_i=0.0224	

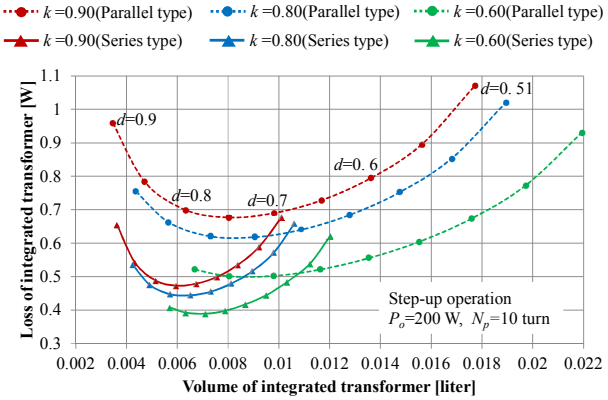


Fig. 8. Relationship between the losses and volume in case of changing the duty ratio d in the step-down operation.

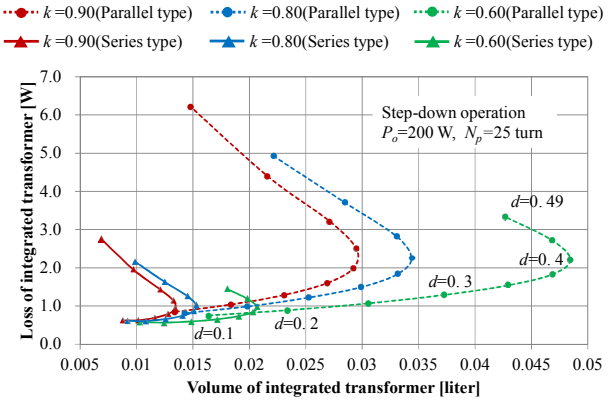
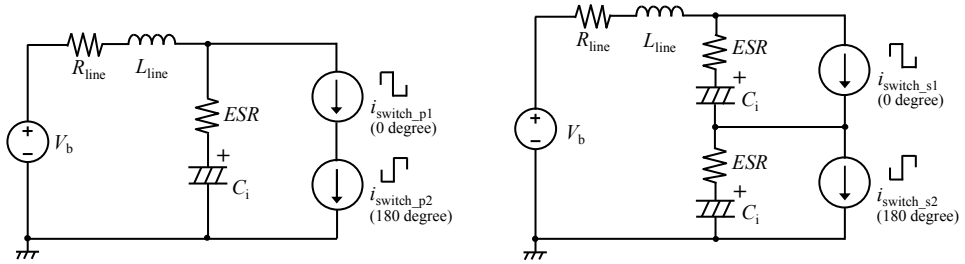


Fig. 9. Relationship between the losses and volume in case of changing the duty ratio d in the step-up operation.

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(a) Parallel type (b) Series type

Fig. 10. Equivalent circuit of the interleaved flyback converter for the input capacitor analysis.

TABLE V. EQUATIONS OF RMS CURRENT AND RIPPLE VOLTAGE FOR INPUT CAPACITOR

		RMS current of the input capacitor I_{rms_c}	Input ripple voltage ΔV_i
Parallel type	d<0.5	$\frac{1}{N_{el}}\cdot\sqrt{\frac{\Delta i_{p_p}^2\cdot d}{6}+I_{c2}^2\cdot\frac{2d}{1-2d}}$ (29)	$\frac{ESR}{N_{el}}\cdot\left(\frac{I_{c2}}{1-2d}+\frac{\Delta i_{p_p}}{2}\right)+\frac{I_{c2}\cdot d\cdot T_s}{C_i\cdot N_{el}}$ (30)
	d>0.5	$\frac{1}{N_{el}}\cdot\sqrt{\frac{\Delta i_{p_p}^2\cdot(3\cdot d-1)}{6}+I_{c1}^2\cdot\frac{2(2\cdot d-1)}{1-d}}$ (31)	$\frac{ESR}{N_{el}}\cdot\left(\frac{I_{c1}}{1-d}+\frac{\Delta i_{p_p}}{2}\right)+\frac{I_{c1}\cdot T_s}{C_i\cdot N_{el}}\cdot(2d-1)$ (32)
Series type	d<0.5	$\frac{1}{N_{el}}\cdot\sqrt{\frac{\Delta i_{p_s}^2\cdot d}{12}+I_{c3}^2\cdot\frac{d}{1-d}}$ (33)	$\frac{ESR}{N_{el}}\cdot\left(\frac{I_{c3}}{1-d}+\frac{\Delta i_{p_s}}{2}\right)+\frac{I_{c3}\cdot d\cdot T_s}{(1-d)\cdot C_i\cdot N_{el}}\cdot(1-2d)$ (34)
	d>0.5	$\frac{1}{N_{el}}\cdot\sqrt{\frac{\Delta i_{p_s}^2\cdot d}{12}+I_{c3}^2\cdot\frac{d}{2(1-d)}}$ (35)	$\frac{ESR}{N_{el}}\cdot\left(\frac{I_{c3}}{1-d}+\frac{\Delta i_{p_s}}{2}\right)+\frac{I_{c3}\cdot T_s}{C_i\cdot N_{el}}\cdot(2d-1)$ (36)

$$\Delta i_{p_p}=\frac{k_i}{2}\cdot(I_i+\beta\cdot I_o) \quad (37) \qquad \Delta i_{p_s}=\frac{k_i}{2}\cdot(2I_i+\beta\cdot I_o) \quad (38)$$

$$I_{c1}=\frac{1}{2}\cdot\left(I_i+\frac{1-2d}{1-d}\cdot\beta\cdot I_o\right) \quad (39) \qquad I_{c2}=\frac{1}{2}\cdot\left(I_i+\frac{1-3d}{1-d}\cdot\beta\cdot I_o\right) \quad (40) \qquad I_{c3}=\frac{1}{2}\cdot\left(2I_i+\frac{1-2d}{1-d}\cdot\beta\cdot I_o\right) \quad (41)$$

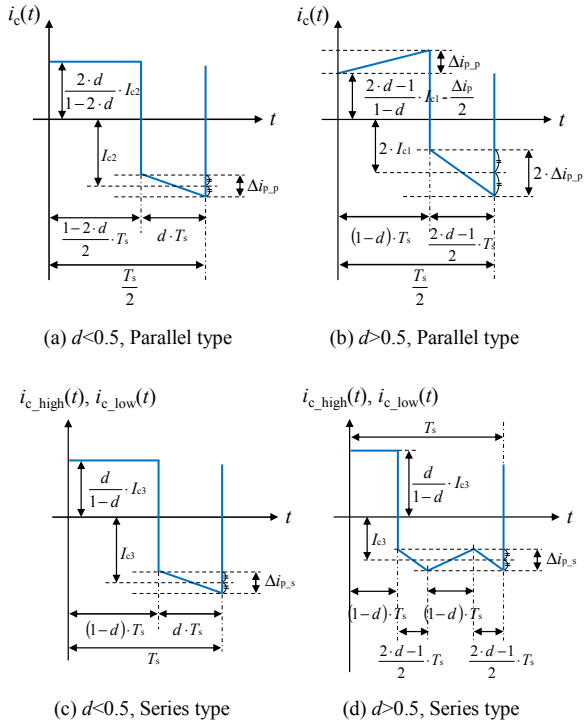
from the viewpoint of high power density. Moreover, from these results, the power losses of the integrated transformer increase when the coupling coefficient k is increased. On the other hand, the power losses of the integrated transformer are decreased when the coupling coefficient k increases. However, the core losses may slightly differ in this situation because the flux density of the integrated transformer is DC biased.

III. ANALYSIS AND DESIGN OF THE INPUT CAPACITOR

Finally, we discuss the design method for the input capacitors of the interleaved flyback converter with integrated transformer. In addition, the RMS current, the ripple voltage, and the volume of the input capacitors are analyzed. Then, we compare the parallel type with the series type about the volume and the power losses.

A. Design method and Analysis

In general, in designing the input capacitor, the RMS current flowing through the capacitor, the ripple voltage between both terminals of the capacitor, and the breakdown voltage are taken into account. Therefore, the current flowing into the capacitor should be analyzed. The current flowing into the capacitor of the interleaved flyback converter can be analyzed from the simplified circuit in Fig. 10. In this paper, it is considered that the equivalent circuit of the capacitor includes not only the capacitance C, but also the Equivalent Series Resistance (ESR). All the components of the primary side placed after the input capacitor can be replaced by a periodically switching current source as shown in Fig. 10. And,

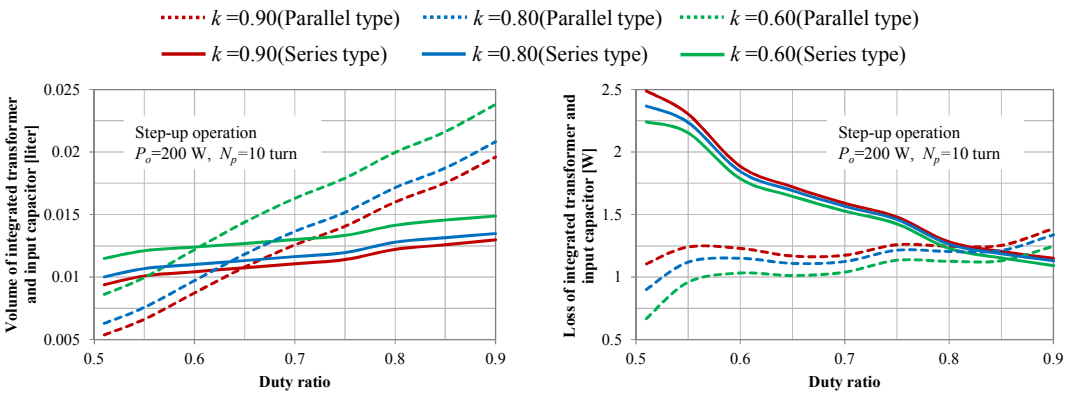


(a) d<0.5, Parallel type (b) d>0.5, Parallel type

(c) d<0.5, Series type (d) d>0.5, Series type

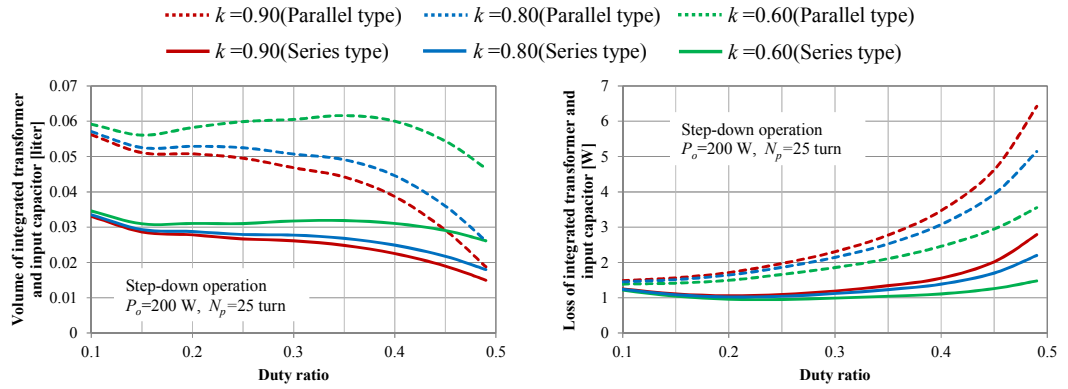
Fig. 11. Waveform of the current flowing through the input capacitor.

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(a) Duty ratio vs. Total volume. (b) Duty ratio vs. Total loss.

Fig. 12. Total volume and losses of the integrated transformer and input capacitor in step-up operation.



(a) Duty ratio vs. Total volume. (b) Duty ratio vs. Total loss.

Fig. 13. Total volume and losses of the integrated transformer and input capacitor in step-down operation.

in this work, we assume the worst-case where all the current ripple is by-passed by the input capacitor (high impedance) [17]. Moreover, the coupling coefficient of the integrated transformer is defined as k=1 for simplicity. From the above, the current waveforms of the each type are shown in Fig. 11. Moreover, the results of the RMS current I_{rms_c} and ripple input voltage ΔV_i are summarized in Table V. Here, N_{el} is the number of capacitors in parallel. The loss caused by the *ESR* P_{ci} is calculated as follows [18]:

$$P_{ci} = ESR \cdot I_{rms_c}^2 \tag{42}$$

Besides, the power losses caused by the leakage current are neglected because they are much smaller than the losses caused by the *ESR*.

B. Comparison of the total volume and power loss

From the above results, we can compare the parallel type with the series type in terms of volume and power losses. The circuit parameters are the same as Table IV. In addition, the ratio of the input ripple voltage $\Delta V_i/V_i$ is determined as lower than 0.01. In case of the step-up operation, it is assumed that the aluminum electrolytic capacitor NHG series (radial-lead type) manufactured by Panasonic is used. In case of the step-down operation, it is assumed that FP series (surface-mounting type) manufactured by Panasonic is used. We select the capacitance and number of capacitors in parallel in order to reduce the volume as much as possible.

First, we set the breakdown voltage of the input capacitor. By considering the voltage margin, we set the breakdown voltage as 35V in the step-up operation of the parallel type, and 16V in the step-up operation of the series type. Moreover, we set the breakdown voltage to 450V in the step-down operation of the parallel type, and 250V in the step-down operation of the series type. Then, we calculate the number of capacitors in parallel N_{el}. N_{el} needed to satisfy the RMS current and the ripple voltage. In fact, N_{el} is determined by the results of Table V.

In this context, in case of the step-up operation, the dependency of the duty ratio on the total volume and the total power losses (sum of the integrated transformer and input capacitor) is shown in Fig. 12. Similarly, the results of the step-down operation are shown in Fig. 13. If the capacitors with high performance (low ESR) are used in the series type, the technique becomes more effective for high power density than the parallel type because the losses of the input capacitor occupy most of the total loss. From the results of the step-down operation (Fig. 13.), the series type can achieve high power density compared to the series type under the all duty ratio.

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IV. CONCLUSION

This paper provides the design guidelines, the electromagnetic analysis, and the quantitative analysis of the volume and power losses of passive components for two types of interleaved flyback converter with integrated transformer. Then, volume and power losses of passive components for the parallel and the series type are compared under the same conditions. In case of the step-down operation, it is concluded that the series type can achieve higher power density compared to the series type. In case of the step-up operation, it is concluded that parallel type is effective for higher power density compared to the series type. However, if capacitors with high performance (low ESR) are used in the series type, the technique becomes more effective for high power density than the parallel type because the losses of the input capacitor are the biggest contributor to the total losses.

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